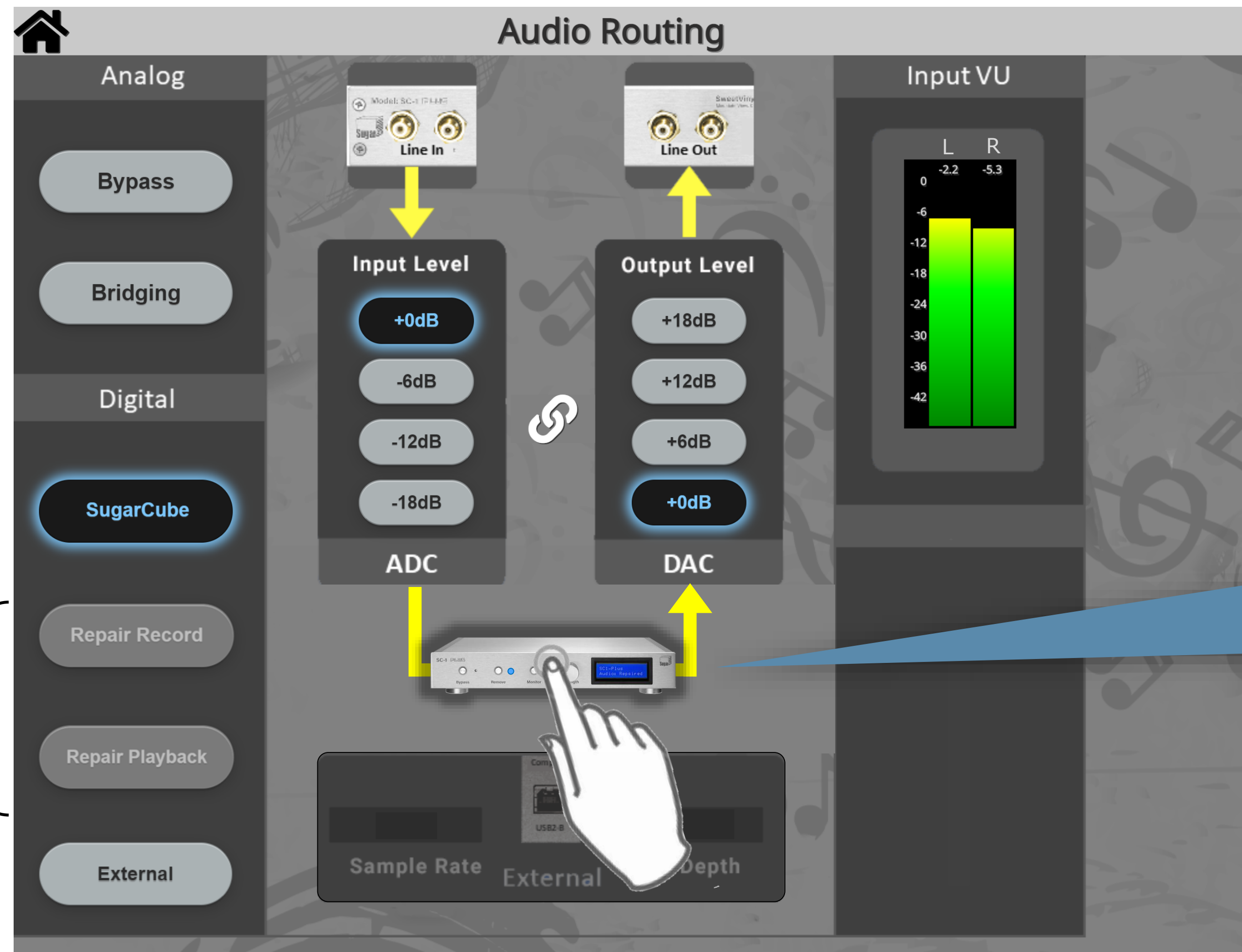


5.15 Audio Routing Menu — SugarCube Mode

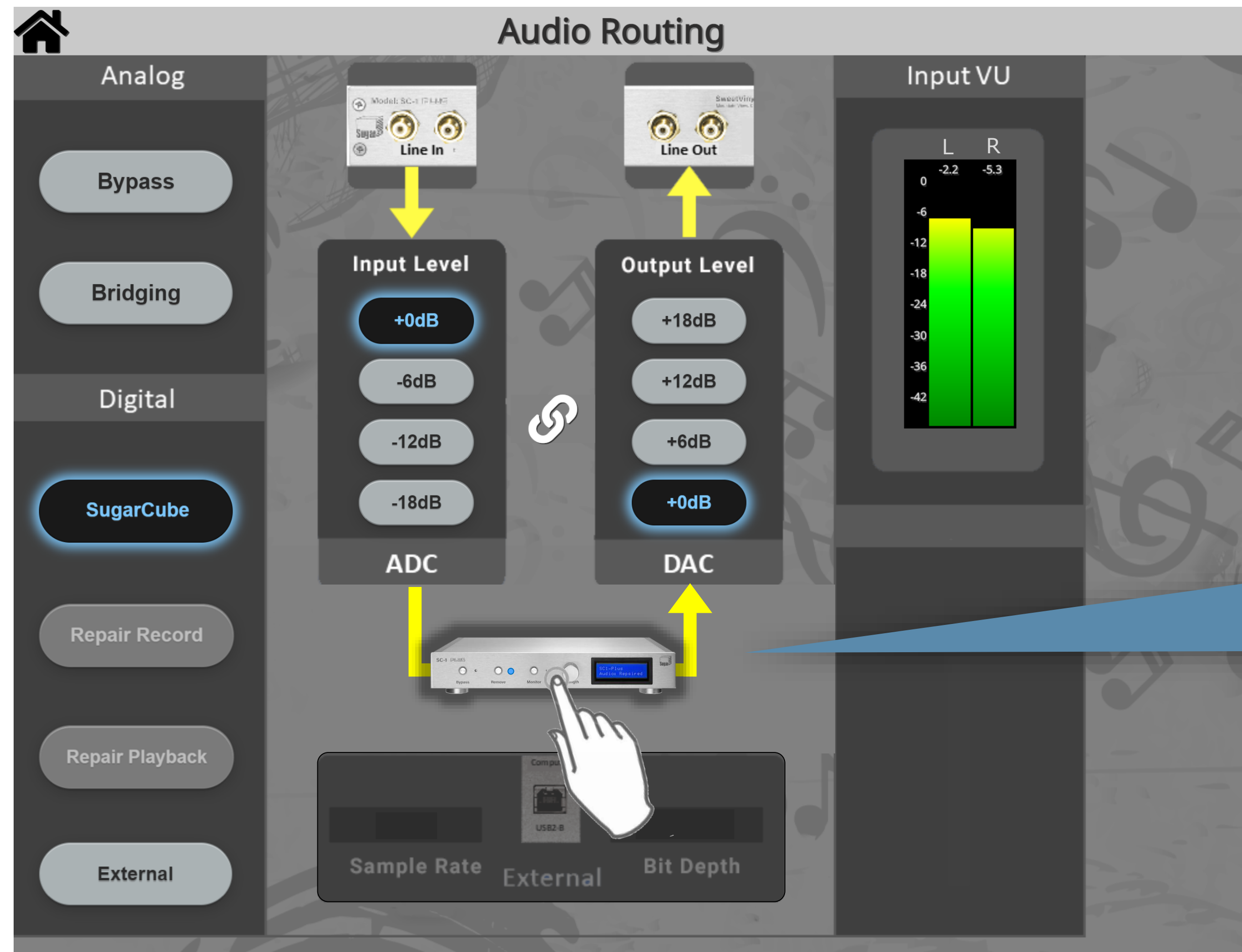


Not available in SugarCube mode, go to 'External' to activate.

Click on the SugarCube **icon** to engage/dis-engage repair functions

In **SugarCube** mode, audio is processed by the SugarCube as standard on all Sugar Cubes. In addition, the **Un-Repaired Digital** signal is passed to the attached computer via the **ADC**.

5.16 Audio Routing Menu — SugarCube Mode



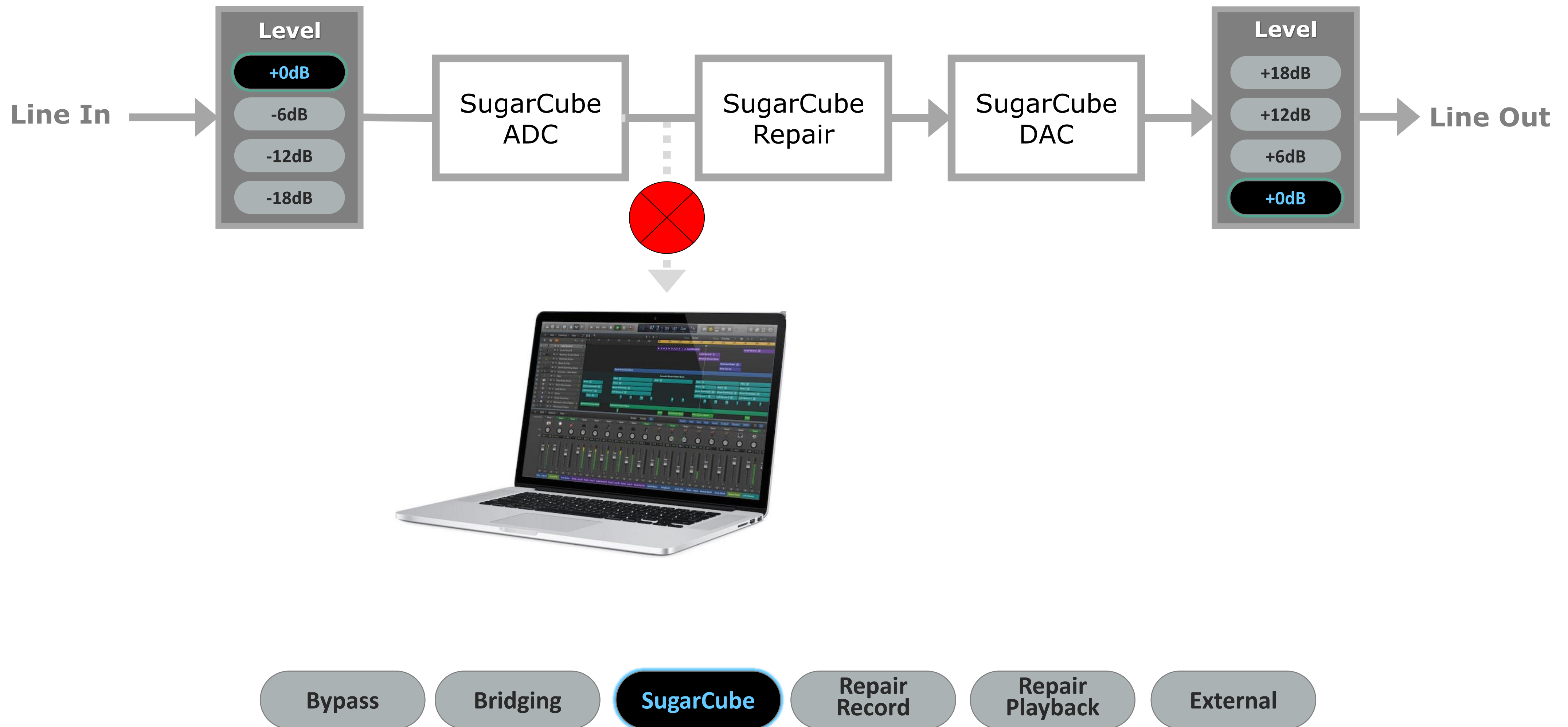
Click on the SugarCube **icon** to engage/dis-engage repair functions

Same as clicking on the front panel 'Remove' button



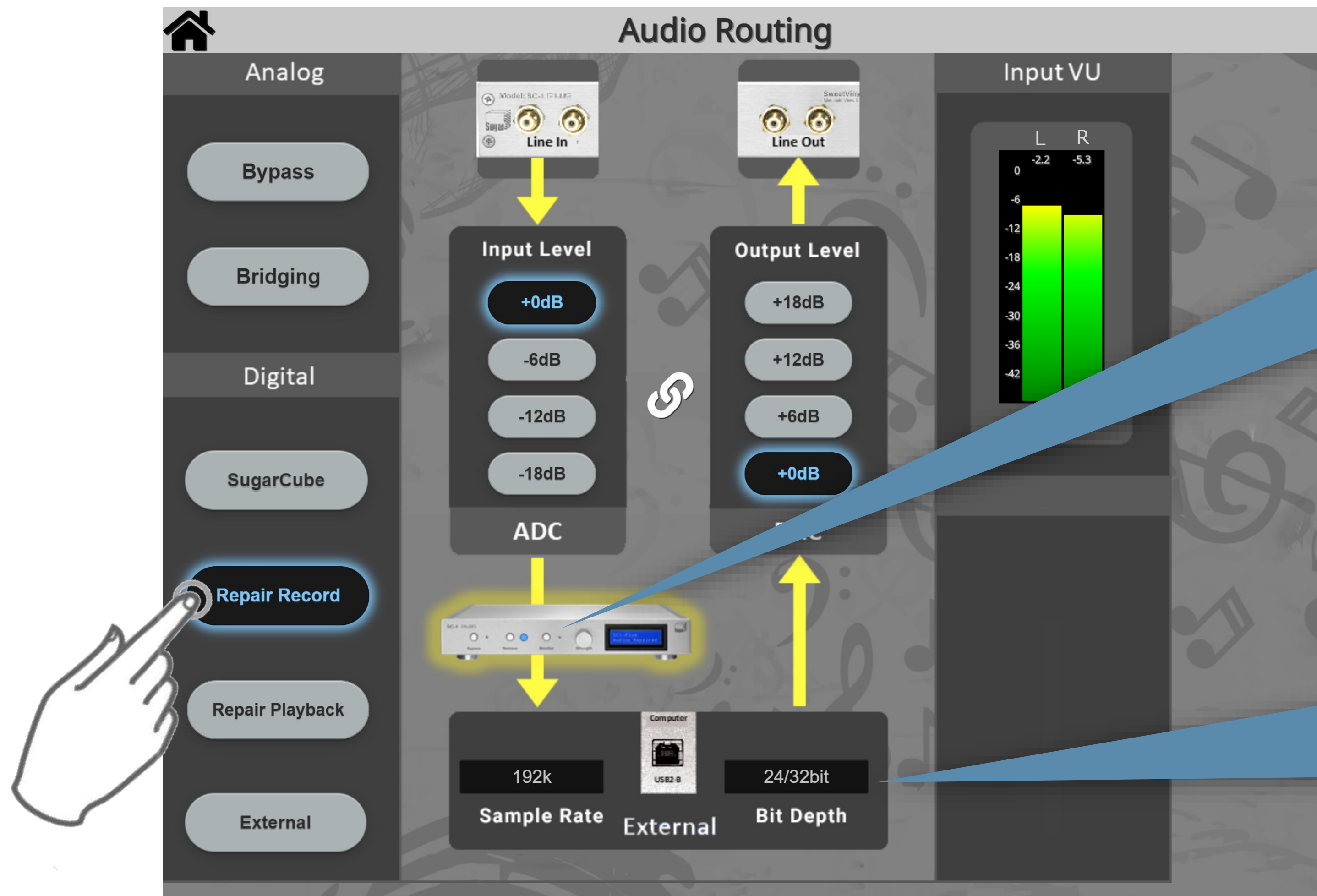
5.17 Logical Flow — SugarCube Mode

Normal SugarCube operations without a digital signal from ADC to the workstation



In **SugarCube** mode, **Line-In** is processed by the SugarCube and audio is repaired and sent to **Line Out** as standard on all SugarCubes.

5.18 Audio Routing — Repair Record Mode



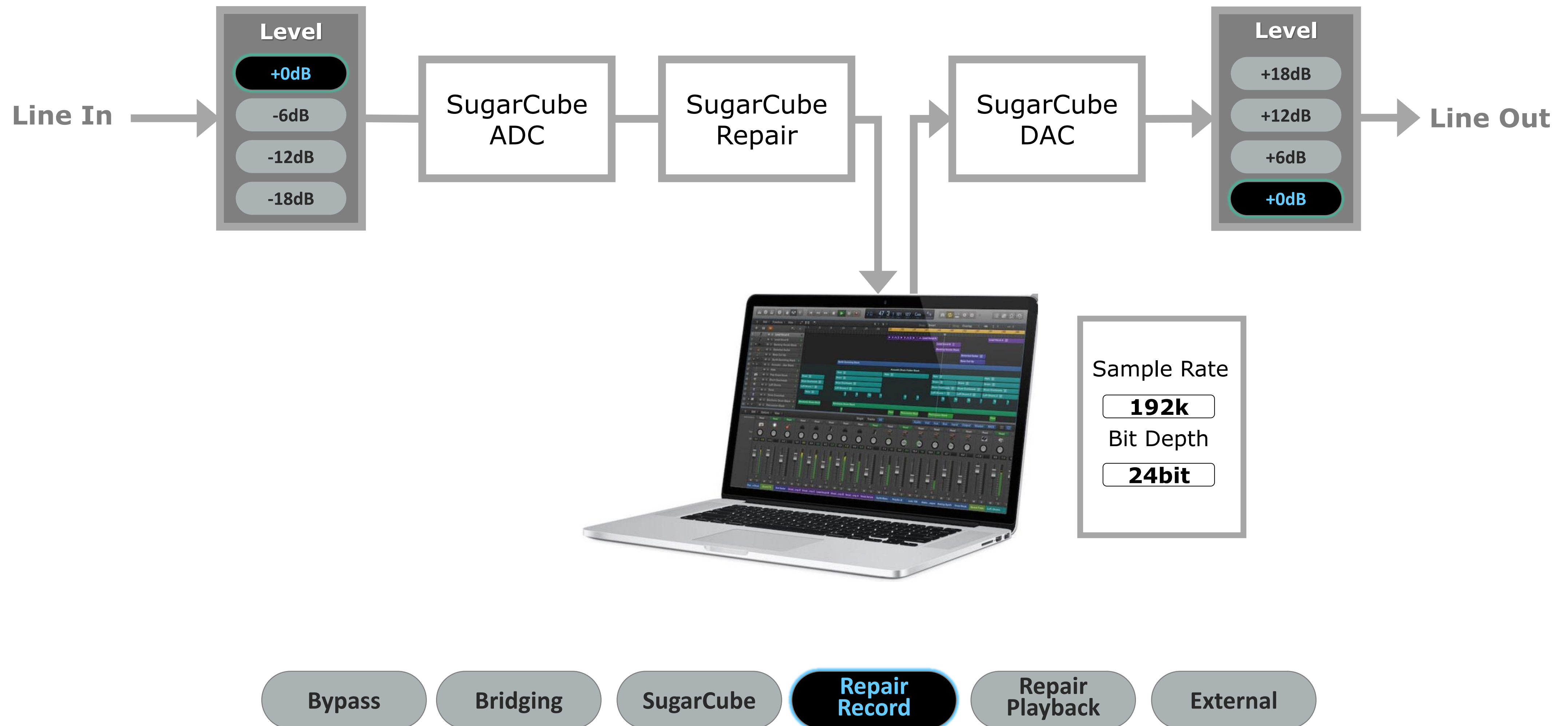
'**Repair Record**' inserts the SugarCube repair processing **after** the **ADC** to repair audio before outputting to the external device

If the connected computer operates at **192k/24bit**, the system will be able to utilize SugarCube repair functions

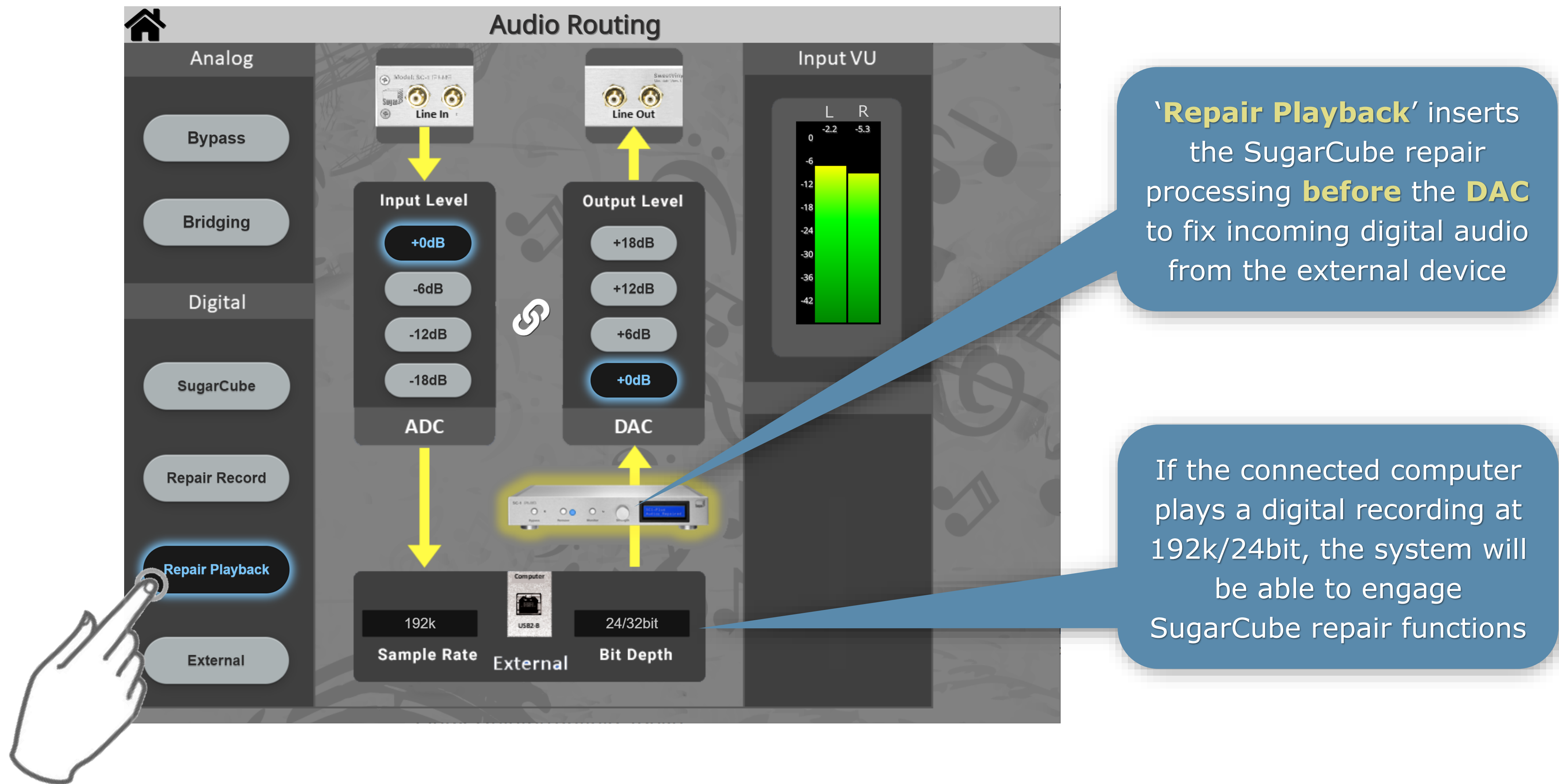
Note: Please **press again** the button of **any mode** of operation (even though it is already selected) if the external device **loses lock** with the SugarCube (you will certainly know due to obvious digital noise.) This resets the internal timing and restores the audio.

5.19 Logical Flow — Repair Record Mode

**SugarCube repairs audio before sending out to workstation.
Workstation directly drives the DAC.**



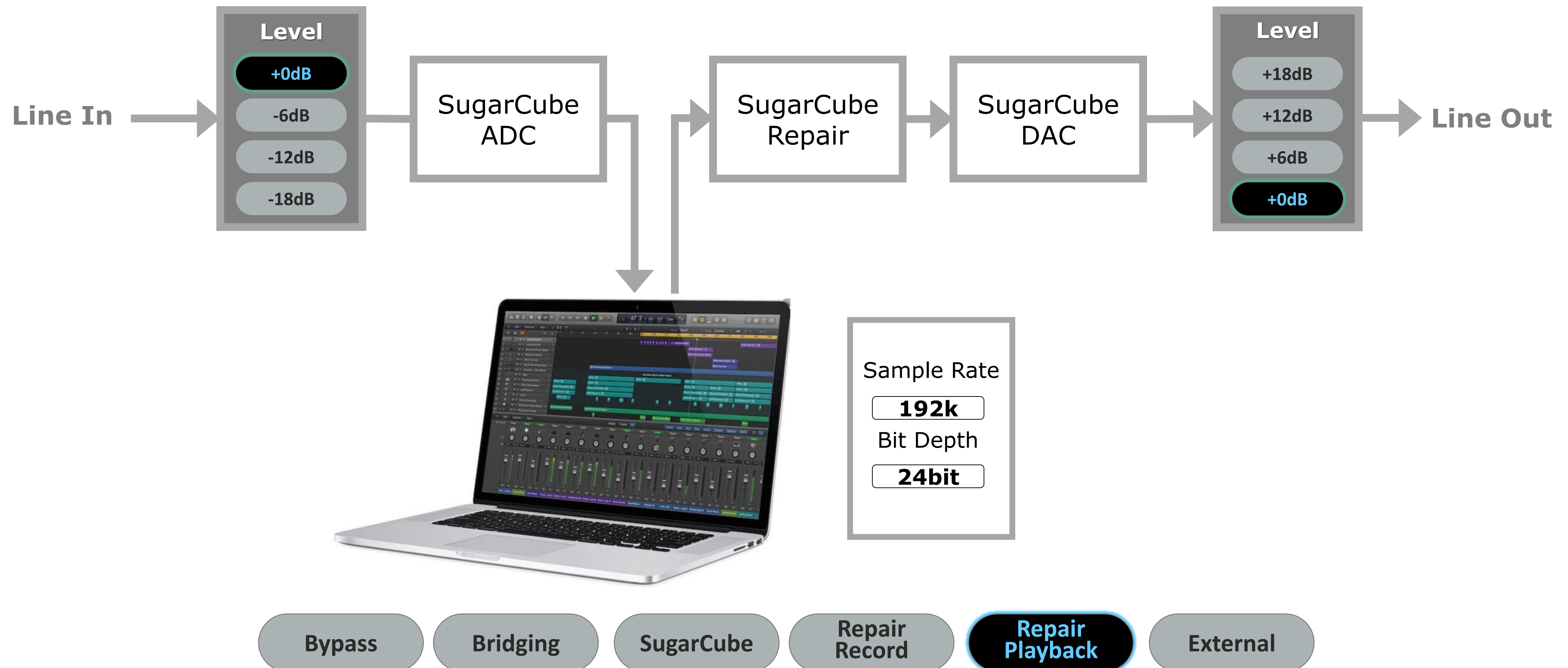
5.20 Audio Routing — Repair Playback Mode



The **Repair Playback** mode is particularly useful allowing you to play any digital file from your library to remove clicks & pops and/or remove tape hiss from previous recordings from any source from the past.

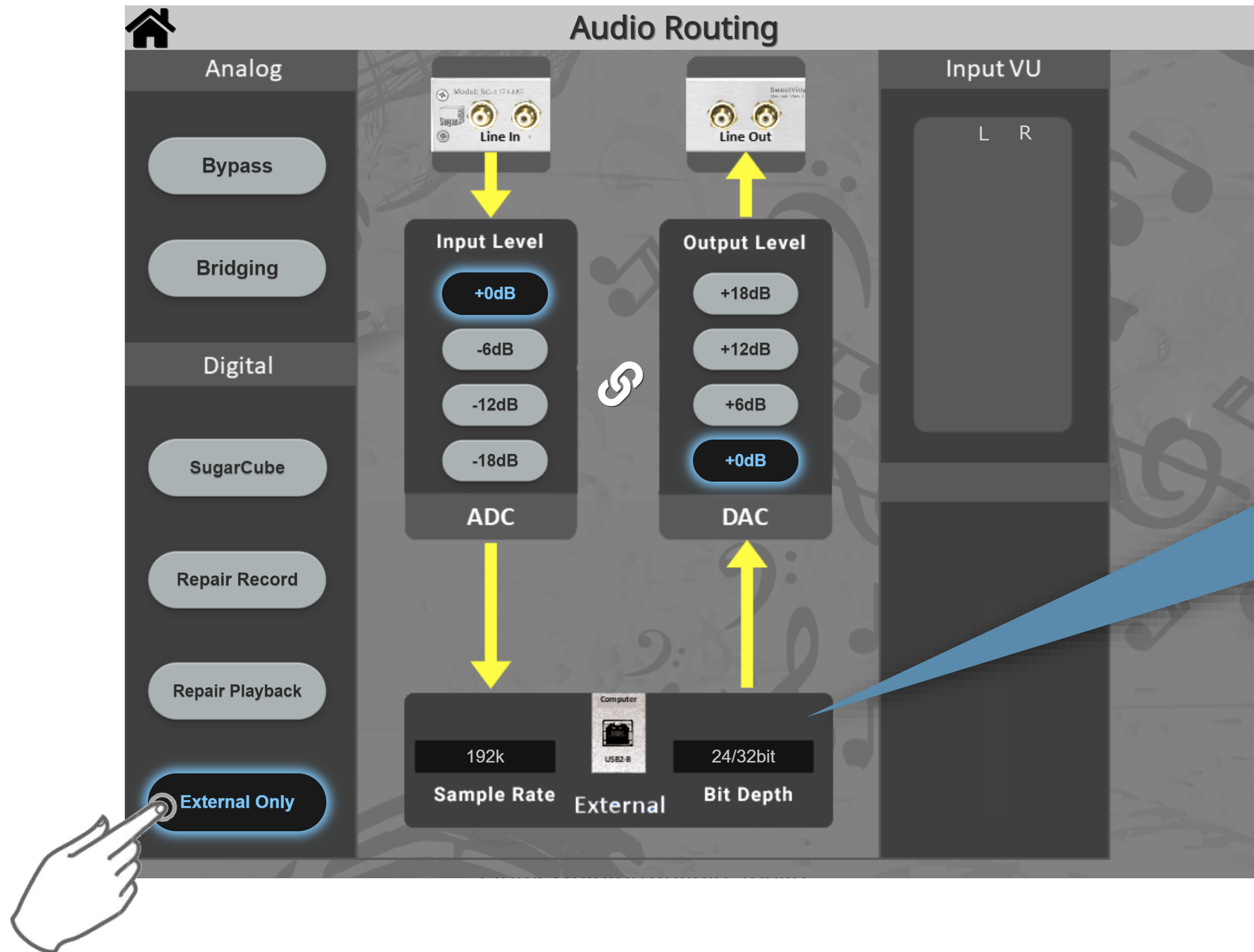
5.21 Logical Flow — Repair Playback Mode

Digitized audio goes directly to workstation. Workstation output feeds SugarCube for repair before DAC.



Repair Playback is our **recommended mode** for operation. All of your files will be **recorded** “in the raw” with no SugarCube processing applied. Then, on **playback**, you can adjust click & pop removal as well as apply noise reduction (if needed) on a recording-by-recording basis or use no processing at all if desired.

5.22 Audio Routing Menu — External Mode

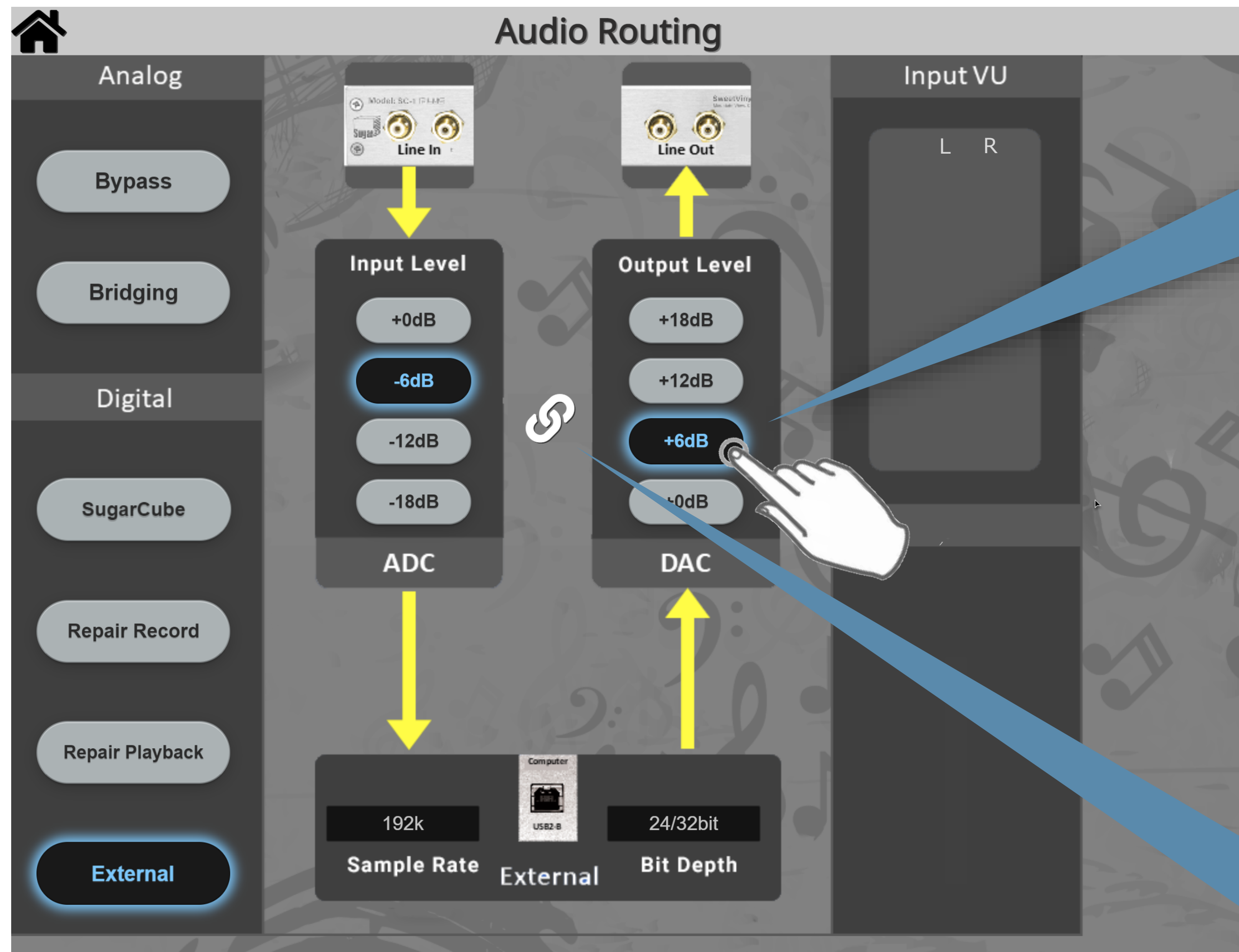


External mode allows an external device to use the **ADC** and **DAC** at **any sample rate**.

In **External** mode, the **SC1-Plus** acts like **any other** studio-quality **ADC/DAC** pair and will function at any standard sample rate and bit depth from 192k down to 44.1k and at 16/24/32 bit sample sizes.

Note: Please **press again** the button of **the current mode** of operation if the external device **loses lock** with the SugarCube (you will certainly know due to obvious digital noise.) This resets the internal timing and restores the audio.

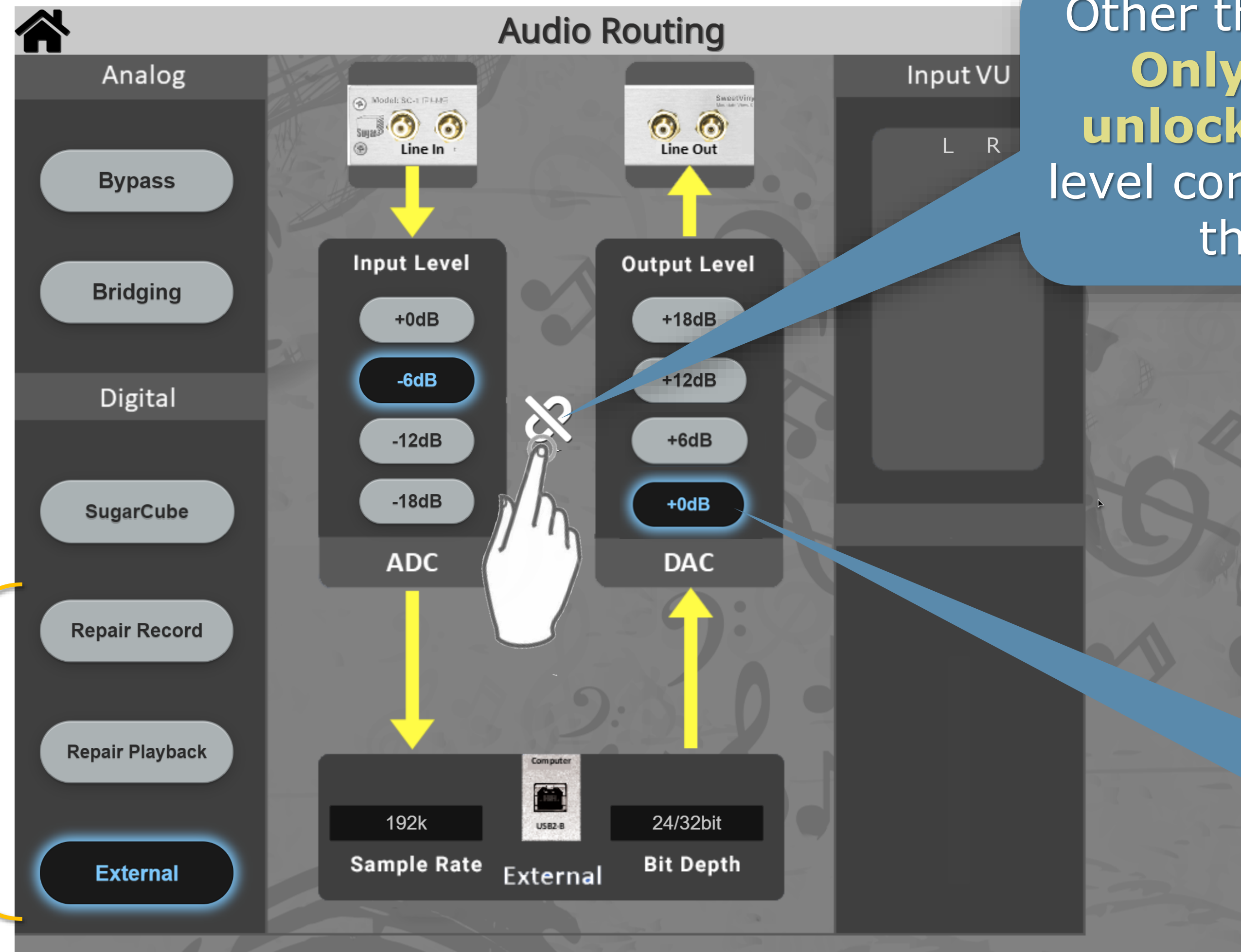
5.23 Audio Routing Menu — External Mode



Output Level can be changed during playback to any of the four available levels.

However, when locked, input level will change in the **opposite** direction.

5.24 Audio Routing Menu — External-Mode Levels

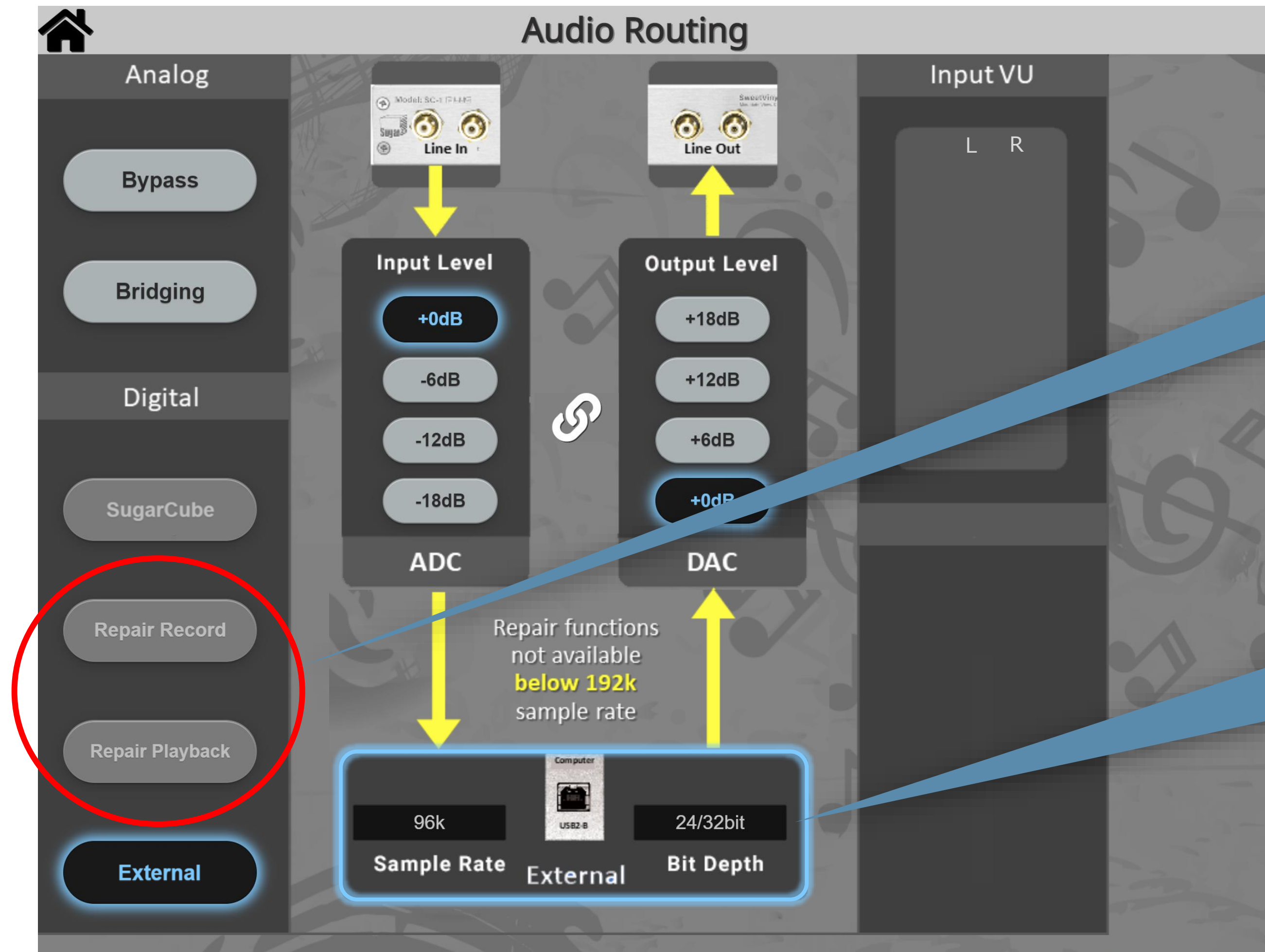


Other than in **SugarCube**
Only mode, you can
unlock input and output
level controls by clicking on
the chain icon.

When unlocked, **Input Level** and **Output Level** can be set independently.

External Modes

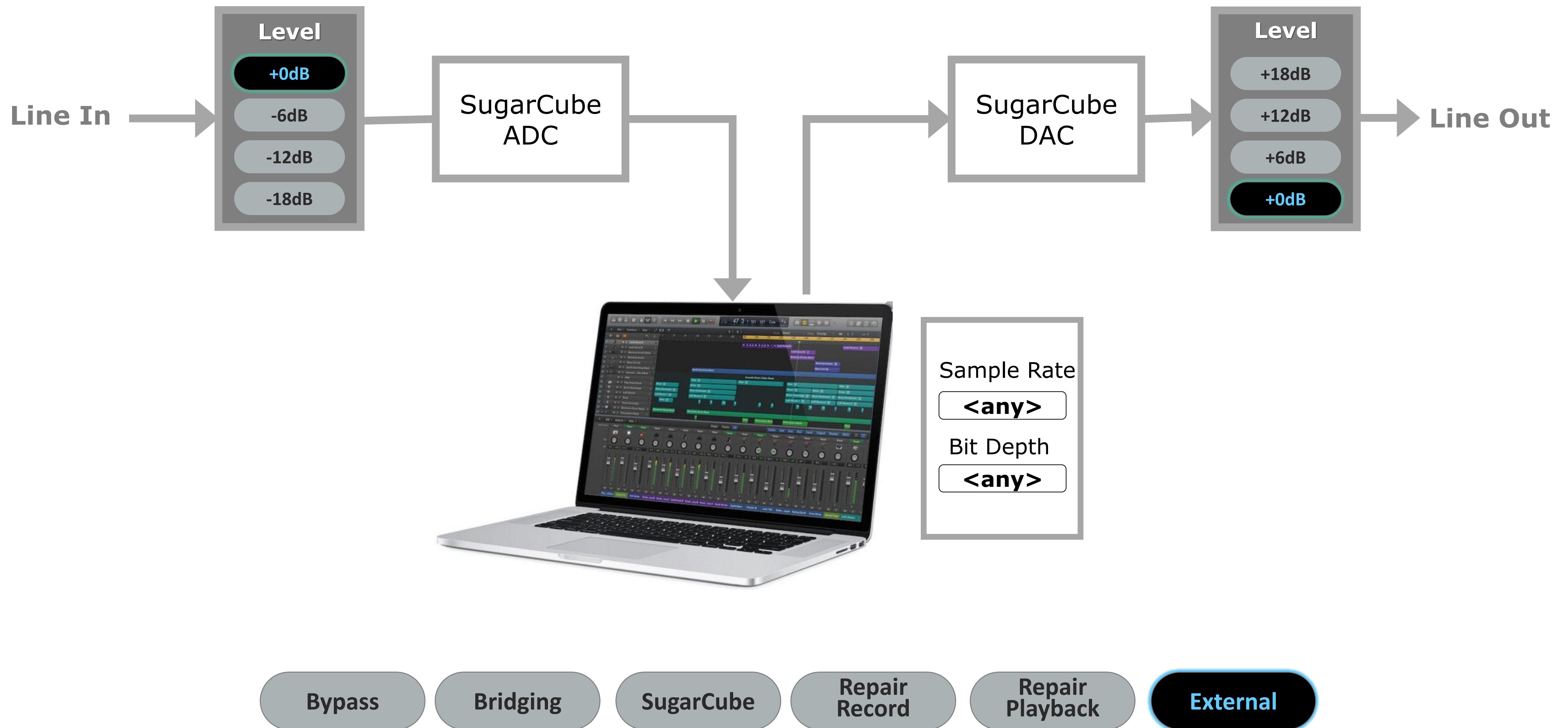
5.25 Audio Routing — External Device **Not** Set to 192k/24bit



In **External** mode, it is fine to operate at **any sample rate and bit-depth below 192k/24bit**, but SugarCube **repair functions** do not operate **below 192k/24bit**.

5.26 Logical Flow — External Mode

**SugarCube repair functions are out of the loop.
Workstation uses ADC & DAC directly**



In **External** mode, the SugarCube will operate at **any sample rate and bit depth** from 192k/24bit down to 44.1/16bit, just like any other studio-quality ADC/DAC pair.

FAQ

Some common questions
& answers about using
the SC-Plus series



Basic connectivity

- **Q: Line-In level setting**, why so many choices?
- **A:** You can select four steps each 6dB apart to accommodate low- to high-level output equipment. Whatever level you select for input is mirror the opposite way for output (i.e.-- choosing -6dB line in results in +6dB applied to line out so line in always equals line out.)
- **Q: What is bridging?**
- **A:** This is an analog routing mode where line in goes directly to line out through the bypass relay just like in Bypass mode but is tapped at a high impedance (about 100k ohms) and passed through the internal analog-to-digital converter and then sent out to whatever is attached to the SC1-Plus digital interface. The attached computer can operate at any sample rate or bit depth in this mode.
- **Q: How do I connect through tape loops?**
- **A:** Tape Out goes to the SugarCube Line In and the SugarCube Line Out goes to Tape In. Sometimes the tape loop is labeled Aux or Rec but is the same thing.
- **Q: How do I connecting an external computer?**
- **A:** The SC1-Plus appears as a standard ADC/DAC to a connected PC, Mac, or music server. No special drivers are needed for any device that can accommodate Class 2 USB audio. Also, we have no drivers for devices that do not support Class 2 USB.
- **Q: My computer only goes to 48k, how to I fix that?**
- **A:** You can always use the SC1-Plus in **External** mode as an ADC/DAC at 48k (or any other sample rate), but you cannot then use the other modes of the Plus. Your external computer will be internally disconnected from the SugarCube when not in **External** mode.

Repair modes

- **Q:** when I use the **noise reduction**, do I have to **start at the lead in groove** or can I play any track on the record?
 - **A:** This is a yes and no answer: for the first time you play any record using noise reduction (NR), you must **start the noise reduction process then drop the needle** on the lead in groove.
 - **A:** Then, if you do not stop the process (that is leave it running), if you are playing, for example, a box set of records from the same album, the “**fingerprint**” of the noise that the SugarCube created stays in memory and, if you don’t turn off the SugarCube, you can drop the needle anywhere or play other side without having to **stop** then **re-start NR** then drop the needle in the lead in groove. The **NR** process will continue to work perfectly.
 - **A:** Furthermore, you can extend the above concept if you have LPs from the same label and era (example: late-1950’s Blue Note), the noise “**fingerprint**” that the SugarCube picked up on the first side of the first LP should work for all other LPs of your Blue Note collection when you use **NR**.
 - **A:** However, if you play an LP from say, EMI, that label used a different vinyl plastic formula in making their LPs so their (EMI’s) noise “**fingerprint**” will be different and will require **stopping** the **NR** process and **re-starting** then dropping the needle once again in the lead in groove.
 - **A:** Finally, you could use **NR** and only one time drop the needle for a “fingerprint” in the lead in groove then never do it again, **NR** will still work but it will not be optimal and there may even be a drop in volume.
 - **A:** Bottom line: **NR** is a very powerful tool that can do an amazing job of lowering noise in recordings up to -20dB. It is well worth your time to **start** and **stop NR** when the LP is **different**.

Repair modes

- **Q:** What does **EQ** do?

- **A:** LPs are made with a process that boosts the highs and diminishes the lows during the cutting of the master disk. This process was fundamental to record making all the way back to Thomas Edison's days in the early 1900's. However, until the mid to late 1950s, each manufacturer (record label) had their own process that used differing amounts of boost and cut. So, an RCA Victor record would not sound the same played on a non-RCA turntable, a standard business practice at the time.
- **A:** More importantly, if you are playing 78s (using a 78-rpm phono cartridge) through your system, your phono pre-amp is almost surely using the RIAA equalization which became fully adopted in the early 1960s. RIAA has a much more drastic cut of high-frequencies and a correspondingly strong boost of the bass frequencies on playback than did any of the 78 recordings made prior to the emergence of the LP in the early 1950s. If you listen to a 78 even with a 78 needle, your phono pre-amp will cut much more treble and correspondingly boost much more bass than needed making the playback sound muddy and lacking in detail.
- **A:** When you apply EQ on the repair page with the correct equalization choice from our menu, you 78 playback will sound remarkably cleaner and balanced and with detail that you did not know was there.
- **A:** Recall that a 78 record is a direct-to-disc recording as the tape recorder had not yet been invented. When you remove the clicks and pops (which are far more pronounced than on vinyl LPs) and then you lower the loud surface noise of a lacquer disc, and then correct the EQ curve, the sound quality is surprisingly delightful.

Audio routing and use with external devices

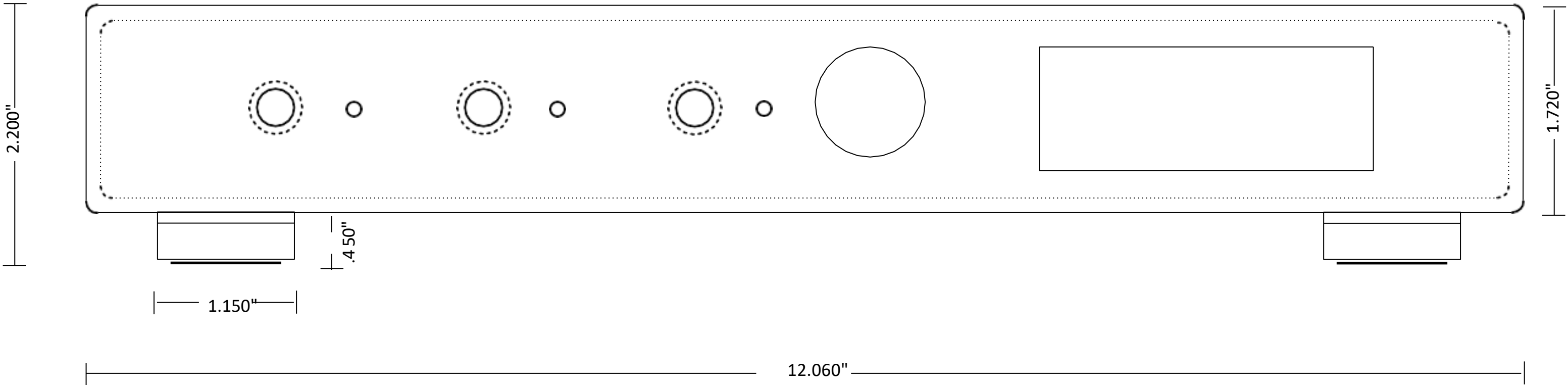
- **Q:** Why can't my computer run **Vinyl Studio** at any sample rate I want?
- **A:** You can run at any sample rate/bit depth that you wish if you are operating your SC1-Plus either in the **Bridging** mode or **External**-only mode. When you want to use the SugarCube processing with external recording, you need to operate at 192k/24bit to match our internal digital processing
- **Q:** Why would you want to remove clicks & pops during playback (**Repair Playback** mode) instead of during recording (**Repair Record** mode?)
- **A:** There are a reasons to operate your SC1-Plus with **Repair Record** as well as with **Repair Playback**. (1) for the **Repair Playback** mode, you can record batches of LPs without any processing and then apply click and pop removal and/or noise reduction on playback on a case-by-case basis just like might you do when you listen to an LP. If you were to use **Repair Record**, you would want to be sure you had all your settings optimal because once recorded, you cannot change the settings. However, if you were making a lot of recordings to save, say, MP3's for your car or mobile phone (i.e., non-critical listening) then it would be best to use **Repair Record** mode.
- **Q:** Why do I only hear noise or nothing at all from my **external DAC** when connected to the Plus except in **SugarCube** mode?
- **A:** The **External DAC** mode is a carryover from our other SugarCube models. In this mode, the **External DAC** is receiving audio from the SugarCube processor inside the Plus. So, in **External** mode, the SugarCube is **disconnected** from the audio pathway and the ADC and DAC are **independent audio units** now connected to just your computer as the graphic shows.
- **A:** Also, the **External DAC** will work in the **Repair Playback** mode because the SugarCube has control of the **DAC** but the not the **Repair Record** mode where the **external computer** has control.

Technical Specs

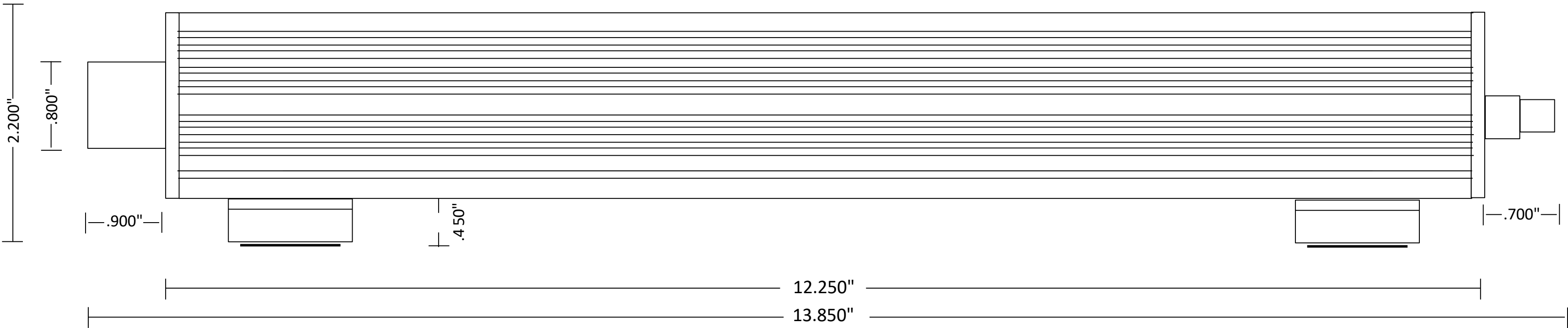


SugarCube SC1-Plus Mechanical:

Front View



Side View



Technical Specifications

- Line In maximum level = +22.7 dBu
(approx. 10.6V rms or 30.0V p-p)
- Line Out level = Line In level
- THD+N = 0.0031% (A-weighted)
- Frequency Response = 25Hz to 60kHz +/-0.1dB
- Frequency Response = 15Hz to 75kHz -3dB
- DC in = 5 volts, center-pin positive
- Average Power Dissipation = 11 watts

General Info

Support

Thank you for purchasing this SugarCube product.

Please contact customer support at: info@sweetvinyl.com

Trademarks

SweetVinyl™ and SugarCube™ are trademarks of SweetVinyl Inc.
Any non-SweetVinyl trademarks are used for reference purposes only.

Compliance

For the current EU Declaration of conformity, visit:
<http://support.sweetvinyl.com/regulatory>

For regulatory compliance information in North America, visit:
<http://support.sweetvinyl.com/regulatory>

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